

**Environment, Social, or Governance? Unpacking the Black Box of what
U.S. Banks prioritize in Loan Contracts**

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Abstract

Do banks adjust the terms of syndicated loans to the ESG performance of their borrowers, and through which contractual channel does this adjustment operate? Drawing on the literature on the pricing of non-financial risk in private debt markets and on a sample of 2,944 loan tranches issued to 589 large publicly listed U.S. firms between 2010 and 2025, five contractual terms (spread, amount, maturity, collateral and covenants) are examined jointly rather than in isolation, and the ESG score is decomposed into its individual pillars. ESG performance is found to be priced only in the spread. Contrary to a substantial body of prior work, no evidence is found that ESG performance is reflected in loan amount, maturity, collateral requirements or the presence of financial covenants once the remaining contractual terms are held constant, which suggests that previously documented effects on non-price terms may partly reflect the omission of this conditioning. Consistently, no evidence is found that the environmental, social and governance pillars are weighed differently by lenders. The direction of an ESG change is further shown to matter only for non-price terms, while ESG disclosure quality was found to be a moderator of the relationship only for the social pillar. No intensification of ESG pricing was observed after the Paris Agreement: loans became uniformly cheaper and less covenant-heavy after 2015, but no additional reward was granted for sustainability performance. Taken together, the results indicate that ESG factors influence credit contracts mainly by affecting the compensation offered to the parties involved, rather than through specific protective clauses in the contracts.

Keywords: ESG performance, Syndicated loans, Loan contract terms, Bank lending, ESG pillars, ESG disclosure quality, Cost of debt.

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1. Introduction

Nowadays ESG considerations are influencing a growing share of business ventures. What began as a purely reputational concern for big corporations has turned, in conditions of tightening regulations and constantly increasing expectations of stakeholders, into a force that shapes all key business decisions: financial, strategic and acquisition ones. Such a shift is especially relevant for financial institutions, and for banks in particular, due to the fact that environmental, social and governance factors are no longer treated by lenders as reputational decoration, but as potential sources of credit risk. Regulatory exposure, litigation, the risk of stranded assets and reputational contagion all have the capacity to affect the ability of the firm-borrower to service its debt, which is exactly the type of information a bank-lender is expected to incorporate into the conditions of the loan it provides. That is why with the increase in ESG concerns for banking institutions it becomes vital to understand whether, and to what extent, ESG risks are actually reflected in the terms on which credit is granted.

This question is particularly important in the context of syndicated loans, which constitute the dominant source of external debt financing for large publicly listed corporations and are the type of loans covered in the current research. Syndicated loans differ from publicly traded debt in one crucial aspect: they are not priced anonymously by the market but are negotiated bilaterally between the borrower and a group of lenders. Moreover, what is being negotiated is not a single price, but a whole package of contractual conditions, which includes the spread, the amount of the loan, its maturity, the requirement of collateral and the presence of financial covenants. All of these terms are determined jointly and can be substituted for one another, meaning that a bank which is not able to charge a higher price to a risky borrower still has the possibility to demand security or to shorten the horizon of the commitment instead. This peculiarity of syndicated lending makes it a uniquely suitable setting for examining how

sustainability considerations enter credit decisions, since it allows us to observe not only whether ESG performance matters, but also through which contractual channel it operates.

The current thesis joins the empirical banking literature that studies the pricing of non-financial risk in private debt markets. This academic conversation was largely initiated by Goss and Roberts (2011), who were among the first to demonstrate that corporate social responsibility is reflected in the cost of bank debt, and has been recently systematized in the critical review of Carnevale and Drago (2024). The conversation itself lies at the intersection of three research streams: the literature on bank loan pricing and contract design (Goss & Roberts, 2011; Bae et al., 2018; Houston & Shan, 2022), the literature on ESG performance and the cost of debt (Cheung, 2020; Cai & He, 2022; Huang et al., 2024), and the emerging literature on the quality and comparability of ESG information itself (Berg et al., 2022; Attig et al., 2025).

Existent research on this topic has already established several findings. First, it is broadly documented that firm-borrowers with stronger ESG performance obtain loans at lower spreads, while poor ESG results and ESG-related incidents are associated with an increase in the cost of borrowing (Goss & Roberts, 2011; Cai & He, 2022; Qian et al., 2023). Second, the effect of ESG is not limited to price terms only, but also extends to non-price contractual conditions, such as the intensity of covenants, the requirement of collateral and the maturity of the loan (Bae et al., 2018; He et al., 2021; Hui et al., 2024). Third, the relationship appears to be asymmetric, in the sense that lenders are found to penalize the deterioration of ESG performance considerably more heavily than they reward its improvement (Becchetti & Manfredonia, 2022; Carnevale & Drago, 2024). Finally, the strength of these effects is conditional on a number of moderating factors, which include the quality of ESG disclosure of the borrower (Attig et al., 2025), the size of the firm and its degree of bank dependence (Houston & Shan, 2022), and the environmental sensitivity of the sector in which the borrower operates (Chen et al., 2021; Huang et al., 2024).

However, despite this accumulated evidence, several important issues remain unresolved. The first and the most significant one is that the majority of existing studies rely on composite ESG scores, which aggregate highly heterogeneous and sometimes conflicting dimensions of sustainability performance into a single figure. Such aggregation is problematic in itself, since ESG ratings provided by different agencies are known to diverge substantially from one another (Berg et al., 2022), but it also leaves the underlying mechanism unobserved. The studies which do decompose ESG into its separate pillars typically conclude that the environmental dimension is the most influential one on the basis of the fact that the environmental coefficient is statistically significant while the others are not. Such reasoning, though, does not actually establish that one pillar is priced more heavily than another, due to the fact that the individual significance of two coefficients says nothing about whether they differ from each other. Establishing relative importance requires a formal test of the equality of coefficients, which the existing literature has rarely performed. The second unresolved issue is that loan terms are predominantly studied one at a time, and overwhelmingly the spread is the term which receives attention. Since the contractual terms are negotiated as a package, examining the spread in isolation makes it impossible to distinguish a situation in which ESG is not priced at all from a situation in which it is priced through a different contractual margin. The third issue concerns the robustness of the documented association itself. Firms with higher ESG scores tend to be larger, less leveraged and more profitable, which are precisely the characteristics that make a borrower attractive to a bank independently of any sustainability considerations. Whether the ESG effect survives once these characteristics are properly controlled therefore remains an open question.

These gaps matter for several groups of actors. For firms, which allocate limited budgets to sustainability initiatives, it is essential to understand which dimension of ESG performance, if any, is actually rewarded by lenders, since investment into social reporting represents a poor

use of resources if banks respond exclusively to environmental indicators. For regulators, who are currently designing and expanding mandatory disclosure regimes, it is important to know whether credit markets reward disclosure at all and which type of disclosure they find credible. And for banks themselves, which increasingly declare the integration of ESG factors into their credit risk assessment, evidence on whether their own contractual behavior is consistent with such declarations is of direct relevance.

The aim of this study is to identify which components of ESG performance, if any, are independently reflected in the contractual terms of U.S. syndicated loans, and through which contractual margins these effects operate.

Closely related to this aim, the following research question is formulated: do U.S. banks price the ESG performance of their borrowers into syndicated loan contracts, and if they do, which ESG dimension and which contractual term does this effect operate through? In order to answer this question comprehensively, it is decomposed into a set of sub-questions, each of which corresponds to one of the hypotheses developed in Chapter 2. These sub-questions ask whether overall ESG performance is associated with loan terms once firm characteristics are held constant (H1); whether lenders react asymmetrically to the deterioration and the improvement of ESG scores (H2); whether the environmental, social and governance pillars are priced differently from one another (H3); whether the quality of ESG disclosure moderates the relationship (H4); and whether the time period, the size of the firm-borrower and the pollution intensity of its sector act as further moderating factors (H5-H7).

The analysis is grounded in several key theories. Signaling theory (Spence, 1973) provides the basic logic of the first hypothesis, since ESG performance can be understood as a costly and observable signal of the otherwise unobservable quality of management and of the long-term orientation of the firm. The theory of information asymmetry and adverse selection (Stiglitz &

Weiss, 1981) underlies the hypothesis on disclosure quality, due to the fact that credible and verifiable reporting reduces the information gap between the firm-borrower and the bank-lender and therefore should strengthen the response of the lender to ESG signals. The risk-management view of corporate social responsibility (Godfrey, 2005) motivates the hypothesis on asymmetry, as it treats ESG performance as a form of insurance-like reputational capital, the loss of which is more consequential than its accumulation. Finally, agency theory and the literature on covenant design (Jensen & Meckling, 1976; Smith & Warner, 1979) justify the decision to examine five contractual terms jointly, since they establish that collateral and covenants function as monitoring substitutes in situations where price alone cannot fully compensate the lender for the risk it bears.

The empirical approach of the current research is based on a sample of syndicated loan tranches issued to publicly listed U.S. firms between 2010 and 2025, for which contract-level information is obtained from the LSEG DealScan database and firm-level ESG scores and financial data from LSEG. The unit of analysis is the loan tranche, and five contractual outcomes are examined: the all-in spread drawn, the amount of the tranche, its maturity, the presence of collateral and the presence of financial covenants. The three continuous outcomes are estimated through Ordinary Least Squares, while the two binary ones are estimated through logistic regression, in both cases with standard errors clustered at the borrower level and with year and two-digit SIC industry fixed effects included, so that identification rests on within-year and within-industry variation. Due to the fact that the five terms are negotiated jointly, each specification additionally controls for the remaining four terms, together with a standard set of firm-level controls for leverage, profitability, interest coverage, size, growth opportunities and asset tangibility, and for whether the facility is sustainability-linked.

The contribution of this thesis is fourfold. First and most importantly, it tests the relative importance of the ESG pillars formally rather than inferring it from patterns of statistical

significance, which represents a methodological correction to a literature that has routinely drawn conclusions about which pillar matters most from evidence that does not support such conclusions. Second, by modelling five contractual terms jointly and controlling for the remaining terms in each specification, the study treats the loan agreement as the negotiated package that it actually is, which makes it possible to identify not only whether ESG performance is reflected in loan contracts but also where in the contract it is reflected. Third, the results indicate that the aggregate ESG score carries considerably less independent information than the univariate evidence suggests, and that the raw ESG discount documented in the literature largely reflects differences in firm size, leverage and profitability rather than sustainability performance as such; this finding constitutes a scope condition on a body of research that has predominantly reported positive associations, and it is consistent with the observation of Carnevale and Drago (2024) that lenders respond to specific, event-based ESG information rather than to composite scores. Fourth, the study covers a recent U.S. sample spanning the transition from LIBOR to SOFR and the emergence of sustainability-linked lending and explicitly controls for the sustainability-linked status of the facility, which ensures that any association detected reflects genuine risk-based pricing rather than the mechanical contractual linkage of ESG targets to loan terms. Taking together, these contributions suggest that the pricing of sustainability risk in bank lending is narrower, and the results do not allow us to conclude that lenders weigh one pillar more than another, than the aggregate-score literature currently implies.

Regarding the structure of the thesis, section 2 develops theory and hypotheses. In section 3 the sample, variables and empirical models are described. Section 4 presents the univariate and regression results. Section 5 discusses the conclusions, limitations of the analysis and future implications.

2. Theory and hypotheses

2.1 ESG Performance and Loan Terms

General definition of ESG is a framework that includes environmental, societal and governance factors, that are usually used by investors to estimate the firms' level of sustainability (Carnevale & Drago, 2024). From the perspective of sustainable banking, ESG is defined as environmental, social and governance performance which is contractually tied to KPIs in SLLs – sustainability-linked loans (Kim et al., 2025). First, the environmental pillar usually includes greenhouse gas emissions, energy and water consumption, management of waste, biodiversity protection, strategy of decreasing climate change and state of preparedness for physical climate risks, such as flooding, hurricanes and other extreme weather events (Corporate Finance Institute, 2026). This pillar captures both: the reduction of negative environmental effects and the proactive adaptation of various sustainable practices (transition to renewable sources of energy, reduction of carbon footprints), all of which are reducing exposure of a firm to regulatory and transition risks (Arzenton, 2020). Second, the social pillar stands for a relationship of a firm with stakeholders, including employees, suppliers, customers and the communities in which it operates (Corporate Finance Institute, 2026). Key dimensions of pillar “S” contain human capital management, fair wages, employee health and safety, diversity, inclusion, supply chain labor standards and community engagement (Corporate Finance Institute, 2026; Arzenton, 2020). Finally, the governance pillar focuses on the internal structures, processes through which a firm is being directed and managed. The “G” pillar includes board composition, independence, alignment of executive compensation stakeholder interests, anti-corruption policies, tax transparency and rights of shareholders (Corporate Finance Institute, 2026). Strong governance is understood as the institutional foundation that makes commitments on pillar “E” and “S” come true, due to the fact that it determines whether sustainability policies

are substantively incorporated in firm strategy or remain superficial disclosures (Arzenton, 2020; Teti et al., 2025).

Existent research offers analysis of the general ESG effects on loan contractual conditions. For instance, research in China shows that borrowers with higher ESG scores receive lower interest rates (Qian et al., 2023). Moreover, good ESG performance also helps companies to get loans with longer maturity. The evidence shows that with decreasing ESG risk, companies improve their chances to extend their repayment timeline (Carnevale & Drago, 2024). While with the rise of ESG risks, expressed in selective disclosure gap (known as greenwashing), borrowers also face more restrictive non-price terms of loan in general including maturity of the loan (Hui et al., 2024). Also, banks positively consider whether the firm-borrower has some percentage of sustainable institutional ownership. In such case, the banks offer lowered AISD¹ and can possibly provide higher loan amount (Li et al., 2025; Nandy & Shao, 2010). Additionally, at the country-level, borrowers operating in countries with higher levels of sustainability tend to benefit from lower loan spreads (Hoepner et al., 2016).

Moreover, worse ESG ratings tend to deteriorate the conditions of loan contract for the borrower. Banks tend to penalize the existence of ESG concerns or violations with higher spreads of the loan (Cai & He, 2022). ESG incidents, that are damaging overall sustainable reputation of the borrower also end up increasing loan spreads (Carnevale & Drago, 2024). Also, poor CSR performance tends to influence covenants of the loan, except for loans with higher spreads (Bae et al., 2018). Poor CSR performance can cause bank to impose much more restrictive covenant terms, which is proved by high association of the ESG incidents index and more severe level of covenant conditions in the loan (He et al., 2021). Moreover, majority of the research proves that poor ESG performance is most likely to reduce the amount of money

¹ AISD (All-In Spread Drawn) – it is the percentage coupon spread over LIBOR plus the annual fee and the upfront fee paid by borrowers for each dollar drawn down. Typically, as a variable it is used to study the price of the loans.

the bank is willing to lend to the borrower (He et al., 2021; Richardson et al., 2023; Nandy & Lodh, 2012). And CSiR (corporate social irresponsibility) specifically causes increased loan costs for the borrowers, showing that direct harmful practices have much more significant negative effect on the loan contractual terms rather than simple ESG underperformance (Becchetti & Manfredonia, 2022).

Based on that fact and all arguments above, the following hypothesis is proposed, which supports the dominated opinion on the relationship between ESG effects and loan contractual terms.

Hypothesis 1. Firms with higher overall ESG scores get better loan conditions (lower spreads, larger amounts, longer maturities), consistent with improved ESG scores being perceived as a signal of lower default and reputational risk.

2.2 Heterogeneous Effects of ESG on Loan Terms

The evidence shows that the relationship between ESG scores and loan contractual conditions is not symmetric: the positive effects of the developed ESG scores are much less significant rather than the negative influence of deteriorated ESG characteristics. Existing research supports the dynamics described above. It states that lenders penalize the poor performance significantly more than they reward good ESG results (Carnevale & Drago, 2024). In particular, banks tend to increase the spreads when the lowering of Corporate Environmental responsibility (CER) of the borrower is observed. According to the research, the opposite is also true, the better the environmental performance of the company, the lower the loan spreads they are getting, however only in the case if the firm has a well-established reputation, meaning that they are credible and are showing positive ESG performance for years (Cai & He, 2022). Thus, the disbalance between the positive and negative ESG effects influence on the loan contractual conditions is observed. The asymmetry of the effects spreads into others loan contractual terms

as well. Strengths of CSR seem to have quite a small influence on the increase of the amount of loan covenants. At the same time, concerns related to the CSR influence the severeness of loan covenants substantially (Bae et al., 2018). Taken together, these findings suggest that reductions on ESG performance have a bigger effect on loan conditions than improvements on ESG, leading to the following hypothesis:

Hypothesis 2. Among ESG dimensions, negative ESG outcomes are more strongly associated with both higher loan spreads and restrictive non-price terms (collateral requirements, covenant intensity) than are positive ESG effects with the improvement of the loan conditions.

2.3 Individual ESG Pillar Effects

Beyond the aggregate effect of ESG, a growing body of research analyses each pillar in isolation.

Regarding the environmental pillar, studies show that the effect of environmental performance in developed markets operates through a variety of different channels. Starting with regulatory risks, firms that are dealing with more demanding environmental regulations tend to get higher interest rates and other contractual loan agreement adjusted on their higher environmental risk (Carnevale & Drago, 2024). Pollution is another channel of effect. Chemical pollution caused by the company-borrowers leads to higher spreads of the loan, as well as high level of carbon emissions (Chen et al., 2021; Kleimeier & Viehs, 2021). It is also stated in the research that lenders increase the spreads and decrease the maturity of the loan for so called brown borrowers that come from low carbon industries (Ho & Wong, 2023). When it comes to positive environmental effects, eco-practices seem to not have a significant effect on the cost of debt (Karaman et al., 2024). Whereas a firm borrower possessing properties that were green-certified

and in general having good environmental performance lean to decreasing of loan spread from the banks (Cheung et al., 2020; Eichholtz et al., 2019).

Moving to the societal pillar, one of its key dimensions is employee satisfaction. Existing research finds the significant negative relationship between it and the loan price (Chi & Chen, 2021). Moreover, it was found that fairness of employee treatment can either reduce the price of the loan or decrease the intensity of financial covenants, if the treatment is fair. This effect is also found to be true in cases when charges in unfair labor practices lead to significant increase in the spread of the loan (Carnevale & Drago, 2024). When it comes to product safety concerns, the relationship between product recall and loan spread is positive. That means, the more products recall company experience due to safety concerns, the higher spread of loan becomes (Zhang, 2021). Thus, satisfaction of employees and conducting labor practices tend to reduce spreads, whereas failures of product safety tend to increase them.

Finally, moving to the governance pillar, we find evidence that underscores its moderate role in the relationship between environmental or social performance and loan terms. For instance, chemical pollution can lead to increase of the loan spread, however this effect is boosted by weak governance (Chen et al., 2021). Moreover, the positive relationships between products recalls and spreads, is stronger for the borrowers with weaker state of governance (Zhang, 2021). Also, strengths of CSR are only able to decrease the spreads of the loans if they are supported by good corporate governance (Bae et al., 2018). Good and efficient governance practices in general are associated with less strict loan contractual conditions for the firms that are exposed to climate risks (Huang et al., 2022). To sum the “G” pillar plays more of a moderating role for the effects of “E” and “S”, due to the fact that good performance in governance makes the firm borrower ESG results more credible for the banks.

Another empirical paper also distinguish the influence of the role of pillars separately. It is suggested that environmental pillar can have a bigger role than social and governance pillar. Some studies show that environmental performance has the greatest influence on loan spreads compared to the social and governance pillars, with CER consistently outweighing CSR in its effect on lending conditions (Cai & He, 2022). Furthermore, the majority of the research on the current topic test environmental effects in particular, signaling that pillar “E” is the most consistently measurable one with the most significant effect on loan pricing (Carnevale & Drago, 2024).

All of the individual effects of each of the pillars described suppose that the effect on the loan conditions varies between pillars and dimensions and that environmental effects have the highest influence, that is why the following hypothesis can be introduced.

Hypothesis 3. The impact of ESG on loan terms differs between pillars: the environmental, social, and governance scores have distinct and possibly asymmetric effects. Environmental factors often play a bigger role than social and governance effects.

2.4 The Role of ESG Information Quality

Some studies prove that ESG information is inconsistent, even with the diversion of ESG definitions, the main problem is that ESG ratings provided by different agencies are not aligned. That implies that it is extremely hard for the banks to base their decisions on conflicting and impartial data (Carnevale & Drago, 2024). Moreover, research suggests that with the higher level of ESG information disclosure of the borrower banks tend to give better loan conditions. Opposite is also found to be true. In case of selective environmental disclosure or, in other words, greenwashing lenders “punish” companies for less credible information with higher fees and higher loan spreads (Attig et al., 2025). Also, Chinese experience shows that voluntary disclosure is even more valuable for the banks than a mandatory one, since it signals genuine

commitment of the borrower and reduces the adverse selection – the information gap between the firm-borrower and the bank-lender (Hui et al., 2024). This issue is reflected by the fact that lenders are more inclined to give better contractual terms for those firms, who show better results in more objective, event-based measures, such as carbon emissions, environmental incidents, etc., rather than base their analysis on composite scores, that combines a lot of different factors into one figure (Carnevale & Drago, 2024). These findings, along with the moderating role of the governance pillar explained in the previous section, build a fundament for the following hypothesis.

Hypothesis 4. The negative association between ESG scores and loan spreads is stronger for borrowers with higher ESG disclosure quality, as more transparent reporting reduces information asymmetry between borrower and lender.

2.5 Moderating Factors: Time Period, Firm Size, and Sector

Several other factors may moderate the relationship between ESG scores and loan conditions. First, time period has a major impact on ESG sensitivity. Some studies show that before 2015, when ESG issues were not as major as they became later, financials markets in general and banks specifically had much less dependency on the state of the ESG factors (Carnevale & Drago, 2024). Time period also influences the technology used for ESG factors estimation. The precision and tools used to measure ESG performance have changed drastically over the last few decades, meaning that rating scores had to be adjusted and changed rapidly and repeatedly, influencing significant banking perception of them and, therefore, their effect on the loan contractual conditions (Carnevale & Drago, 2024; Kempa et al., 2021). Moreover, some research justifies that the financial environment of the given period actually shapes the way in which ESG is evaluated for the firm in general. Cominotti and Franzini (2022), in their analysis of European syndicated loans spanning 2002–2021, provided direct empirical evidence for this moderation. In the 2009-2014, post financial crisis period, country-level sustainability was

found to moderate significantly the relationship between ESG firm rating and loan spread. It suggested that lenders became more sensitive to ESG signals than they used to be before straight after the serious shock of 2008 financial crisis, when banks were more concerned about borrower reliability and visibility of the risks they might cause. The relationship strengthens further after the adoption of the Paris Agreement in 2015 and the early stages of development of the EU Sustainable Finance Action Plan. The relationship between ESG scores and loan amounts became statistically significant, suggesting that after 2015 ESG has become more relevant in banking lending conditions, however only for above-average ESG borrowers. Such a periodic divide suggests that lenders price and structure loans for ESG-performing firms depend on the time period due to different conditions of the economy these periods provide. Taken together, these findings confirm that time period has a moderating force in the ESG-loan contractual terms relationship and let to develop the following hypothesis.

Hypothesis 5. The relationship between ESG performance and loan contractual terms is moderated by the time period.

Additionally, ESG data availability and visibility of ESG scores in general are firm size dependent. Smaller firms are found to have much weaker influence on their environmental performance on the conditions of their loans (Wellalage & Kumar, 2021; Zhang, 2021). However, it is not the result of small firms being more sustainable, but the result of the banks lacking ESG data on these firms, as a result they make less objective and more risky decisions (Carnevale & Drago, 2024). Moreover, firm size influences the negotiating power of the firm and the level of dependency on the bank. Large firms are not dependent on the bank, they can access several lenders at the same time, public bond markets, etc., which gives them valuable counter power over banks. Whereas the research indicates the opposite for the smaller firms. Due to the lack of resources and less business influence, small firms are more bank dependent, meaning that they have much less negotiating power in the loan contractual terms (Houston &

Shan, 2022). Moderating effects of size are also shown to be significant as some research states that ESG score of two different firms-borrowers can be practically the same, however they will get highly different loan conditions from the bank-lender base on the differences that come from their size. If a firm is large with a good level of investment-grade and spends significant resources on ESG, that indicates for the bank that the firm is capable of credible commitment and managing risk in long term, giving the lender a good reason to give this borrower a loan on better conditions. Otherwise, if the firm is small and spends huge amounts on ESG improving their ESG scores, but is highly levered, for the lender it signals that they should be cautious with borrowing such firm money. Usually, banks review ESG results of such firm with suspicion and give them higher spreads due to higher risk that comes from the firm size (Goss & Roberts, 2011; Bae et al., 2018). This reasoning leads to the prediction that size of the borrower's firm is having a moderating effect on the ESG scores-loan terms relationships, which is reflected in the following hypothesis.

Hypothesis 6. The relationship between ESG performance and loan contractual terms is moderated by the size of the firm.

Finally, the sector of the borrower firm implies a specific level of environmental risk, since the highest correlation between environmental scores and loan conditions is observed in high-polluting and environmentally sensitive industries (Chen et al., 2021; Carnevale & Drago, 2024). Therefore, environmental exposure on a sector level is a key moderator of ESG ratings influence on loan contractual terms due to the fact that banks do not apply ESG pricing uniformly across all industries. Huang et al. (2024) used a comprehensive sample of syndicated loan contracts to demonstrate that banks systematically offer more favorable terms (lower spreads, longer maturities, fewer number of restrictive covenants) to firms-borrower firms with strong ESG performance. Crucially, these benefits were significantly more pronounced for the firms which were operating in high-polluting industries. The study also shows that markets

themselves reward banks for giving preferential lending terms to high ESG-rated borrowers in environmentally sensitive sectors, supporting the practice at an institutional level. The reason for such mechanism existing is that in carbon-intensive industries (energy, heavy manufacturing) poor ESG performance strongly translates into regulatory exposure, increased asset risk and reputational concerns (Huang et al., 2024). Thus, all of the potential consequences are translated into higher loan pricing by the lenders. The evidence described above provides strong ground for the following hypothesis.

Hypothesis 7. The impact of ESG performance on loan contractual terms is heterogeneous across borrower sectors, being more pronounced for firms in high-polluting or environmentally sensitive industries.

3. Sample and Methodology

3.1 Sample

The unit of analysis in the current research is the loan tranche. Sample details on spreads, amounts, maturities, collateral, covenants, loan purpose, syndication structure and lender characteristics come from the Dealscan Database. While ESG data comes from the LSEG database, which provides overall ESG scores and firm financials used to build our control variables (size, leverage, profitability, market-to-book, tangibility, etc.). The sample contains 2,944 loan tranches of 589 publicly listed U.S. firms between 2010 and 2025. To exclude possible influence of extreme values (outliers), all continuous variables are winsorized at the 1st and 99th percentiles via Stata software. In case it does not solve the issue of outliers present the variables are winsorized with 5% and 95%. Not all tranches are included in the multivariate analysis. Out of the 2,944 tranches in the sample, 262 are missing a reported maturity, two do not have a lender count, and 18 lack at least one firm-level control. Additionally, one observation falls into a two-digit SIC code cell with only one loan, which is absorbed by the industry fixed effects. As a result, the estimation sample is reduced to 2,661 tranches.

3.2 Models

3.2.1 Dependent variables

The dependent variables are loan spread, loan amount, maturity, collateral and covenants (see Appendix, table A1). *Loan spread* is measured as the natural logarithm of the all-in spread drawn in basis points over the benchmark rate (LIBOR/SOFR), formally computed as $\ln(\text{allinspreaddrawnbps})$. This represents the total cost of borrowing. This way of calculating is one of the most widely used proxy for the price of credit risk in syndicated loan research. The second dependent variable is *loan amount*, defined as the natural logarithm of the committed tranche amount in millions of USD, computed as $\ln(\text{trancheamountm})$. This variable serves as

a proxy for credit access. *Loan maturity* is measured in months from the date of signing the contract till the expiry date (tenormaturitymonth). It reflects the willingness of the lender to commit funds over a longer horizon. The logarithmic transformation applied to spread, and amount reduces skewness to the right and mitigates the influence of outliers. As these three dependent variables are continuous, they are estimated through Ordinary Least Squares (OLS) regression, with standard errors clustered at the borrower's level, industry and year fixed effects included.

Collateral is a binary variable constructed as $\text{Collateral} = 1$ if $\text{secured} = \text{"yes"}$, and 0 otherwise; where a secured loan signals that the lender perceives higher borrower risk and requires asset protection. *Covenants* are analogically coded as $\text{Covenants} = 1$ if financial covenants are present in the loan agreement ($\text{covenants} = \text{"yes"}$), and 0 otherwise. Both binary variables are estimated through the logistic regression with standard errors clustered at the borrower level, and industry fixed effect, and year fixed effects.

3.2.2 Modeling

Before explaining main explanatory variables, the models for the hypotheses are going to be introduced.

For *H1*, the explanatory variable which is used here is ESG:

$$Y_{it} = \alpha + \beta_1 \text{ESG}_{it} + \gamma' X_{it} + \varepsilon_{it} \quad (1)$$

ESG score is a composite measure sourced from LSEG dataset that combines environmental, social, and governance performance into a single score with values from 0 to 100. Firms with higher values indicate stronger overall sustainability performance

At the firm level, a set of control variables is included. *Leverage* is measured as total debt divided by total assets (td/ta). It controls the level of financial stability, due to the fact that

the firms which are more leveraged are facing greater risk of default, therefore are expected to get worse spread conditions. *Return on assets (ROA)*, computed as net income divided by total assets ($\text{net_income}/ta$), captures the profitability of the firm: more profitable firms show stronger debt repayment capacity, and such effect is expected to reduce the cost of the debt for the borrower. *Interest coverage* is defined as earnings before interest and taxes divided by interest expenses ($\text{ebit}/\text{interest}$). It measures the ability of the firm to service its existing debt obligations from EBIT, where lower values indicate higher risk of financial distress. *Firm size* is assessed as the natural logarithm of total assets ($\ln(ta)$). It is included to account for the fact that larger firms typically get greater bargaining power in negotiations of loan conditions. Moreover, they tend to have broader access to capital markets and higher levels of financial transparency. All of these factors reduce the cost for them. *Market-to-book ratio (mtb)* proxies for growth opportunities. Firms with higher expected future value could be treated differently by lenders in their assessments of loan conditions. Finally, *tangibility* is computed as property, plant and equipment divided by total assets (ppe/ta). It captures the share of hard physical assets available as potential collateral. This reduces lender exposure and in general is associated with better loan conditions.

At the loan level, *loan size* is estimated as the natural logarithm of the committed tranche amount in millions of USD ($\ln(\text{trancheamountm})$). This variable controls economies of scale in pricing of the loan, due to the fact that bigger loans could carry lower spreads per unit. And when loan size is used as control variable it captures the fact that bigger loans are cheaper and are needed to be held constant. In all models all loan variables are included, except when one of them is used as dependent variable. Moreover, a *sustainability-linked loan (SLL) dummy* is included. It is coded to be 1 if the loan has at least one key performance indicator linked to ESG ($\text{numberofkpis} > 0$) and 0 in other cases. This control variable ensures that any reduction in

loan spreads associated with ESG scores reflects true pricing benefits. Not only capturing loans that are contractually linked to ESG performance targets.

For H2:

$$Y_{it} = \alpha + \beta_1 |\Delta ESG_{it}| + \beta_2 (|\Delta ESG_{it}| \times D_{inc,it}) + \gamma' X_{it} + \varepsilon_{it} \quad (2)$$

The explanatory variable is the change of ESG (in absolute value) and an interaction term (change of ESG X dummy), where dummy equals 1 if ESG increases.

The change in ESG (ΔESG) captures the yearly improvement or deterioration of a firm's ESG score. The ESG Increase Dummy is a binary variable that shows the direction of the change which is stated by the previous variable. If ESG improves it gets value 1 and if not, then 0. Taken together, these two variables make it possible to conduct a test of whether bank-lenders respond differently to ESG improvements versus deteriorations.

For H3:

$$Y_{it} = \alpha + \beta_1 E_{it} + \beta_2 S_{it} + \beta_3 G_{it} + \gamma' X_{it} + \varepsilon_{it} \quad (3)$$

There are 3 main explanatory variables in the same model: E, S and G. Due to the fact that individual significance of two coefficients says nothing about whether they differ from one another, the relative importance of the pillars cannot be established from the significance stars alone. Therefore, the three pillars are estimated jointly. Next, each model with Wald tests of the linear restrictions $\beta_E = \beta_S$, $\beta_E = \beta_G$ and $\beta_S = \beta_G$ are checked, together with a test of their joint equality – a test that checks if all three pillars carry the same weight. Since the pillars differ in dispersion, the model is estimated on standardized scores, so each coefficient measures the effect of a one standard deviation increase and the magnitudes are directly comparable. The signed difference $\beta_E - \beta_S$ is reported alongside the coefficients so that the direction of any gap is visible.

Each pillar of the ESG framework consists of a variety of dimensions. Pillar “E” includes emissions (emissions, waste, biodiversity and environmental management systems), innovation (product innovation and green revenues/R&D/CapEx) and resource use (water, energy, sustainable packaging and environmental supply chain). Pillar “S” constitutes workforce, which includes diversity and inclusion, career development and training, working conditions, and health and safety. Moreover, the social pillar includes human rights, community and product responsibility (responsible marketing, product quality and data privacy). As for the “G” pillar, it stands for CSR strategy (including ESG reporting and transparency), management structure (independence, diversity, committees), compensation, shareholder rights and takeover defenses (Refinitiv, 2022).

For H4:

$$Y_{it} = \alpha + \beta_1 E_{it} + \beta_2 D_{CSR,it} + \beta_3 (E_{it} \times D_{CSR,it}) + \gamma' X_{it} + \varepsilon_{it} \quad (4a)$$

$$Y_{it} = \alpha + \beta_1 S_{it} + \beta_2 D_{CSR,it} + \beta_3 (S_{it} \times D_{CSR,it}) + \gamma' X_{it} + \varepsilon_{it} \quad (4b)$$

In model 4a the E is included, and in 4b the S. The interactions term is included in both models where the dummy equals 1 if CSR is higher than the median. The CSR transparency dummies moderating variable equals 1 if a firm's LSEG CSR strategy score tops the annual sample median and 0 otherwise.

For H5, ESG is the main explanatory variable, interacted with a time-period dummy:

$$Y_{it} = \alpha + \beta_1 ESG_{it} + \beta_2 D_{2015,it} + \beta_3 (ESG_{it} \times D_{2015,it}) + \gamma' X_{it} + \varepsilon_{it} \quad (5)$$

The time period dummy is a binary variable equal to 1 for the post-Paris Agreement period (2015 onwards) and 0 for the earlier period (2010–2014), marking the point after which ESG considerations became more prominent for banks. When it is combined with ESG scores ($ESG \times D^{\text{period}}$), it captures whether bank sensitivity to ESG shifted after 2015. Year fixed effects are taken out of consideration in the analysis to avoid collinearity.

For H6, ESG is the main explanatory variable with a firm-size dummy:

$$Y_{it} = \alpha + \beta_1 ESG_{it} + \beta_2 D_{size,it} + \beta_3 (ESG_{it} \times D_{size,it}) + \gamma'X_{it} + \varepsilon_{it} \quad (6)$$

The firm-size dummy equals 1 if a firm's size (which measured as the natural logarithm of total assets, $\ln(ta)$) is above the annual sample median, and 0 otherwise. The interaction of this moderating variable with ESG ($ESG \times D^{size}$) tests whether larger firms get a bigger spread discount per unit of ESG improvement.

For H7, ESG is interacting with high-polluting-sector dummy:

$$Y_{it} = \alpha + \beta_1 ESG_{it} + \beta_2 D_{poll,it} + \beta_3 (ESG_{it} \times D_{poll,it}) + \gamma'X_{it} + \varepsilon_{it} \quad (7)$$

The high-polluting industry dummy equals 1 for high-polluters (firms in energy sector, utilities, chemicals and heavy manufacturing sectors, which are identified via SIC codes) and 0 in other cases. The interaction of this moderating variable with ESG ($ESG \times D^{highpollute}$) conducts a test that shows if ESG is priced more severely in higher polluting industries. Sector fixed effects are taken out of consideration in the analysis to avoid collinearity.

3.3 Methodology

All of the seven models that are presented in Section 3.2.2 are estimated separately according to the functions that were developed for each of them. The estimator used depends on the nature of the dependent variable. *Loan spread*, *loan amount* and *maturity* are continuous variables; therefore, they are estimated through Ordinary Least Squares (OLS). *Collateral* and *covenants*, in turn, only take values of 0 or 1, that is why logistic regression is used instead. In logit models, the number of observations is lower than the number of observations in OLS models because the cells where the outcome does not vary are automatically dropped.

In all specifications, standard errors are clustered at the borrower level. This is due to the fact that the same firm-borrower frequently appears in the sample several times, with different loan

tranches across different years. Observations coming from the same borrower are therefore unlikely to be fully independent from each other, and not accounting for this within-firm correlation would understate the standard errors, leading to an overstatement of the statistical significance of the ESG coefficient. Clustering is applied at the borrower identifier rather than the firm identifier. In our sample, 32 firms borrow under more than one identifier, usually a parent company and a separately incorporated subsidiary. Therefore, the 589 firms correspond to 622 borrowing entities. This is why the number of clusters reported in the regression tables is higher than the number of firms mentioned in Section 3.1.

Moreover, all models include year and industry fixed effects, included as dummy variables. Industry fixed effects are defined at the 2-digit SIC level rather than the full 4-digit code, in order to avoid creating industry cells with too few observations (singleton dummies), which would otherwise be dropped from the estimation. Year fixed effects absorb time-specific shocks that are common to all borrowers in a given year, such as shifts in monetary policy or in the benchmark rate, while industry fixed effects absorb the unobserved heterogeneity that exists across sectors, capturing the fact that firms from different industries face different baseline levels of credit risk and capital intensity. Including both ensures that the ESG coefficient captures within-year, within-industry variation, and it is not simply driven by broad macroeconomic trends or by the sector composition of the sample. Models for hypothesis 5 and 7 are exception from this pattern as the period dummy in model 5 is collinear with the year effects and the sector dummy in model 7 with the industry effects, so in each case the affected set of fixed effects is dropped.

Since loan spread, amount, maturity, collateral and covenants are loan terms that are jointly determined and typically negotiated together within the same contract, each model also includes the remaining four loan terms as additional controls, together with the firm-level controls described in Section 3.2.2 (leverage, ROA, interest coverage, firm size, market-to-book and

tangibility), the sustainability-linked loan (SLL) dummy, two controls of loan structure, the logarithm of one plus the number of lenders and the logarithm of one plus the number of ESG-linked performance indicators.

All results are estimated in Stata and reported with the `estout/esttab` package, which allows to report, for each model, the coefficients, the clustered standard errors, the significance stars, the number of observations, the R^2 for the OLS models, and the number of firm clusters together with linear restrictions are described below.

As for hypothesis 1, it is tested directly on the coefficient β_1 of model 1. This coefficient is expected to be negative for spread, collateral, covenants and positive for amount and maturity.

Moving to hypothesis 2, it is tested by allowing the effect of an ESG change to differ depending on whether the score went up or down. The absolute size of the change is interacted with a dummy for improvements, but the dummy is not included on its own, so that both lines pass through the same point when the score does not move at all. β_1 is then the effect of a deterioration and $\beta_1 + \beta_2$ the effect of an improvement. Since the hypothesis concerns magnitudes rather than slopes differing, three restrictions are tested after each model: $\beta_1 + \beta_2 = 0$, $\beta_2 = 0$, and the symmetry restriction $2\beta_1 + \beta_2 = 0$, whose rejection is the actual evidence for the hypothesis. The ESG change is built at the firm-year level from consecutive years only, which makes this sample smaller than the others, and for the two binary outcomes average marginal effects are reported by direction, since a logit coefficient is not a probability.

Hypothesis 3 combines all the three pillars, as they are strongly correlated and estimating them separately would let each absorb the others. Individual significance does not establish that one pillar matters more, that is why Wald tests of the pairwise restrictions $\beta_E = \beta_S$, $\beta_E = \beta_G$ and $\beta_S = \beta_G$, and of their joint equality, are estimated after each model. The signed difference $\beta_E - \beta_S$ is reported, thus the direction is visible. Because the pillars differ in dispersion, the

comparison of magnitudes is based on a standardized specification in which each coefficient measures the effect of a one standard deviation increase.

Talking about hypothesis 4, it is tested with two specifications, model 4a based on E pillar and model 4b based on pillar S, each interacting with a dummy for a CSR strategy score above the annual median. The annual rather than pooled median is used because these scores rise over the sample period, so a pooled median would not be entirely accurate. Due to the fact that the hypothesis concerns high-disclosure borrowers, the sum $\beta_1 + \beta_3$ is recovered by a linear restriction test and reported with the coefficients.

As for hypothesis 5, it interacts ESG with a post-2015 dummy. Year fixed effects are dropped, since the dummy is an exact linear combination of them. The period dummy consequently absorbs all differences between the two eras, but the interaction still identifies whether the ESG slope changed. $\beta_1 + \beta_3$ gives the post-Paris association and is recovered by a restriction test.

Hypothesis 6 interacts ESG with a dummy for above-median firm size, again defined annually. Both sets of fixed effects are retained here, and the continuous size control is kept, so the interaction tests whether the ESG gradient differs across size groups rather than whether large firms simply obtain better terms. $\beta_1 + \beta_3$ is reported for above-median borrowers.

Talking about Hypothesis 7, it interacts with ESG with a high-polluting sector dummy defined from SIC codes. The industry fixed effects are dropped, because the polluting-sector dummy is defined from the SIC codes and cannot be separated from them. The ESG association within polluting sectors is given by $\beta_1 + \beta_3$. The model is re-estimated with the environmental pillar, holding the sample fixed to observations available in both specifications and rescaling by in-sample standard deviations, since the pillar is more dispersed than the composite score.

The tables describing Explanatory , control, moderating variables and fixed effects can be found in Appendix, tables A2-A5.

4. Results

4.1 Preliminary analyses

The preliminary step before the multivariate regression analysis is conducted by a univariate comparison of loan contractual terms that helps to provide preliminary conclusions if there is an association between ESG performance and different loan conditions, with no control for specific firm or loan characteristics.

Before conducting univariate analyses, the sample must be analyzed for possible outliers. After which descriptives statistics analysis of the sample was performed. The results are displayed in table 1 below.

Table 1. Summary of descriptive statistics

	Mean	SD	Min	Max
Ln spread	5.063	0.478	3.912	6.438
Ln amount	6.571	1.133	3.418	9.393
Tenormaturitymonth	47.102	22.324	6.000	85.000
Esg	52.789	19.585	10.980	87.700
Escore	45.377	27.929	0.000	92.640
Sscore	54.005	22.169	9.790	93.970
Gscore	57.429	20.932	10.880	93.370
Emissions score	48.046	32.018	0.000	99.080
Resource use	50.646	33.985	0.000	99.150
Env innovation	31.042	32.218	0.000	96.590
Workforce	53.629	26.209	5.170	99.070
Hr	41.974	34.470	0.000	95.970
Community	70.755	20.608	17.150	99.350
Product resp	50.689	28.947	0.000	98.120
Management	59.329	26.137	4.420	99.250
Shareholders	57.521	26.293	2.780	99.210
Csr	47.708	35.338	0.000	99.020
Tdta	0.355	0.166	0.002	0.833
Netincometa	0.046	0.060	-0.173	0.241
Ebitinterest_adj	-0.049	0.247	-1.024	0.252
Mtb	3.511	3.917	-3.360	15.340
Ppeta	0.295	0.238	0.016	0.880
Ln ta	16.123	1.398	12.827	19.415
Number of KPIs	1.857	1.069	1.000	3.000
Number of Lenders	11.911	8.003	1.000	54.000
Ln(Maturity in months)	3.666	0.698	1.792	4.443
Ln(Number of KPIs)	0.471	0.587	0.000	1.099
Ln(Number of Lenders)	2.206	0.823	0.000	3.989

Source: own elaboration based on data from Refinitiv and DealScan

The correlation matrix of all of the variables can be found in Appendix, tables A6-A7.

The sample is divided into two groups based on the annual median ESG score: firms with ESG scores strictly above the annual median are classified as High ESG and those at or below the median are classified as Low ESG. The High ESG group comprises 1,449 observations and the Low ESG group 1,495 observations.

For each of the five dependent variables: loan spread, loan amount, maturity, collateral, and covenants - the group means are estimated, and the statistical significance of the difference is assessed using a two-sample t-test with unequal variances. The results are presented in Table 2.

Table 2. Difference of Means Tests: High vs. Low ESG Firms

	Spread	Amount	Maturity	Collateral	Covenants
High ESG	4.881	6.968	42.182	0.187	0.451
Low ESG	5.221	6.186	53.223	0.377	0.421
Difference	-0.340	0.782	-11.041	-0.190	0.030
P-value	0.000	0.000	0.000	0.000	0.102
T-statistics	-18.172	19.230	-12.487	-11.747	1.637

Source: own elaboration based on data from Refinitiv and DealScan

The results reveal significant differences in loan terms between the two groups across most dimensions. High ESG firms obtain loans at lower spreads. On average 0.340 log points lower than low ESG firms. This difference is statistically significant at the 1% level ($t = -18.172$, $p < 0.001$), providing preliminary evidence consistent with Hypothesis 1. Similarly, high ESG firms are receiving larger loan amounts (0.781 log points, $p < 0.001$). Such dynamics would be a justification for banks providing greater credit access to firms with stronger sustainability performance. The collateral requirements are also shown to be lower among high ESG firms

(18.7% versus 37.7%, $p < 0.001$). This suggests that the lenders believe high ESG borrowers to be less risky.

However, two results from this analysis deserve closer attention and explanation. Firstly, high ESG firms tend to have shorter loan maturities on average (42.2 versus 53.2 months, $p < 0.001$), which is opposite to what H1 predicts. This result could reflect the structural differences between the two groups. This will be examined more thoroughly in the multivariate analysis taking into account controls for firm and loan characteristics. Secondly, the difference in covenants between the two groups is not statistically significant ($p = 0.102$). This would suggest that covenants are less susceptible to ESG performance than price and collateral terms.

Overall, the univariate results are broadly consistent with the prediction that higher ESG performance is associated with more favorable loan conditions, particularly in terms of spread, loan amount and collateral. These findings provide a vital primary validation of the main hypotheses, though more precise interpretation requires the multivariate framework to be developed in one of the following sections.

4.2 Regression results

Starting with H1, the coefficients on the firm's ESG score from model (1) are estimated separately for each of the five loan terms, controlling for firm characteristics, the remaining loan terms, the SLL dummy, and year and industry fixed effects, with standard errors clustered by borrower (Table 3).

Across all five specifications, the coefficient on ESG is statistically distinguishable from zero at conventional levels only for the spread dimension.

Thus, these results do not provide full support for Hypothesis 1 once firm-level controls, the other loan terms, and year and industry fixed effects are accounted for. This stands in contrast

with the univariate evidence found in Section 4.1, where High and Low ESG firms differed sharply and significantly on almost every loan term. The most likely explanation for this is that the raw ESG differences documented in Table 2 are actually driven by factors that are correlated with both ESG performance and loan pricing - in particular firm size, leverage and profitability - rather than by ESG itself. Larger, less leveraged and more profitable firms tend to score higher on ESG and, independently of that, are exactly the type of borrowers that banks are inclined to price more favorably. Once these characteristics are held constant, together with the other loan terms and the industry/year heterogeneity, the aggregate ESG score seems to carry little independent information for the loan terms in this sample. This is broadly in line with prior evidence stating that lenders respond more to specific ESG information rather than to composite scores that bundle a lot of different dimensions together (Carnevale & Drago, 2024), which also motivates the pillar- and category-level tests proposed in Hypothesis 3.

Table 3. H1 - Overall ESG score and loan terms

	Spread	Amount	Maturity	Collateral	Covenants
esg	-0.00315*** (0.000700)	-0.000537 (0.00138)	-0.00750 (0.0310)	-0.00383 (0.00786)	-0.000846 (0.00473)
Loan amount	0.00933 (0.0103)		0.0333 (0.549)	-0.0917 (0.107)	-0.109 (0.0689)
Maturity	0.00199*** (0.000501)	0.0000674 (0.00111)		0.0673*** (0.00685)	-0.00440 (0.00282)
Collateral	0.349*** (0.0300)	-0.0246 (0.0626)	14.57*** (1.312)		0.649*** (0.205)
financial covenants	-0.0354** (0.0180)	-0.0596 (0.0405)	-1.323 (0.864)	0.959*** (0.228)	
present					
Leverage	0.541*** (0.0712)	0.319** (0.145)	-8.822*** (2.744)	1.070 (0.636)	-0.709 (0.456)
ROA	-1.370*** (0.234)	1.516*** (0.324)	2.793 (8.436)	-1.467 (1.770)	2.243* (1.161)
Interest coverage	0.0148 (0.0122)	-0.0336 (0.0229)	0.0701 (0.638)	0.0905 (0.139)	0.0641 (0.0829)

Firm size	-0.0857*** (0.0126)	0.559*** (0.0232)	-5.359*** (0.610)	-0.0779 (0.131)	-0.0734 (0.0844)
Market-to-book	-0.00936*** (0.00240)	0.00938* (0.00492)	0.301*** (0.106)	-0.0681*** (0.0256)	0.0158 (0.0168)
Tangibility	-0.0306 (0.0726)	-0.346** (0.150)	5.765** (2.922)	-0.205 (0.610)	0.648 (0.453)
ESG-linked KPI	-1.136*** (0.249)	0.974*** (0.296)	6.625 (14.61)	28.53*** (1.314)	11.98*** (2.312)
Ln(1+Number of KPIs)	1.232*** (0.196)	-0.982*** (0.275)	1.863 (11.05)	-35.21 (0.752)	-18.09*** (2.094)
Ln(1+Number of lenders)	-0.0196 (0.0158)	0.232*** (0.0370)	9.249*** (0.754)	-0.137 (0.170)	0.818*** (0.108)
Ln(All-in spread drawn, bps)		0.0550 (0.0614)	5.799*** (1.486)	3.572*** (0.291)	-0.383** (0.175)
Constant	6.886*** (0.198)	-2.996*** (0.621)	65.32*** (13.48)	-21.51*** (2.712)	2.862 (1.756)
Observations	2661	2661	2661	2568	2635
R-squared	0.545	0.544	0.395		
Borrower clusters	610	610	610	589	601
Year effects	YES	YES	YES	YES	YES
Industry effects	YES	YES	YES	YES	YES

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

In table 4 the results for Hypothesis 2 are presented.

Table 4. H2 - Asymmetric effect of ESG: improvements vs deteriorations

	Spread	Amount	Maturity	Collateral	Covenants
Change in ESG score	-0.00671 (0.00677)	-0.00680 (0.0191)	-0.516 (0.347)	0.226* (0.118)	-0.100 (0.0635)
Change in ESG score x Increase	0.00776 (0.00696)	-0.00849 (0.0161)	0.760** (0.356)	-0.514*** (0.175)	0.0184 (0.0554)
Improvement effect b1+b2	0.001	-0.015	0.244	-0.288	-0.082
SE of b1+b2	0.005	0.013	0.265	0.088	0.040
p-value b1+b2=0				0.001	
p-value b2=0				0.003	
arms differ					

p-value 2b1+b2=0 symmetry				0.587	
AME deterioration				0.0058	-0.0181
AME improvement				-0.0065	-0.0149
Observations	491	491	491	392	451
R-squared	0.687	0.546	0.485		
Pseudo R-squared				0.812	0.220
Borrow clusters	185	185	185	150	164

Note: Standard errors in parentheses; b1 is the effect of a one-point ESG deterioration. b1+b2 is the effect of a one-point ESG improvement, from a linear restriction test. Columns 4-5 are logit, with average marginal effects in probability points. Year and 2-digit SIC fixed effects and full controls included. SE clustered by borrower.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

Table 4 shows the results of the tests on whether loan terms have the asymmetric dependency on ESG improvements and deteriorations (linear restriction tests for all five contractual terms for H2 can be found in Appendix, table A8). The specification separates the two parts: the absolute ESG change and absolute change interacted with an "increase" dummy. Starting from the first two loan characteristics, neither spread nor amount of the loan have any response to movements of ESG in any direction. Collateral and maturity are the only cases where there is a clean difference for lenders depending on direction. However, the decrease in security requirements with the ESG improvement is essentially of the same magnitude as the increases of the requirements that follows the deterioration of ESG position. When there is a decrease in ESG the probability of collateral increases (0.226), and in case of ESG increase, the probability of collateral decreases ($0.226 - 0.514 = -0.288$, statistically different from zero). As for maturity, deteriorations have no detectable effect (-0.516, SE 0.347), while improvements lengthen maturity by 0.76 months per ESG point ($p < 0.05$). Since the first coefficient is zero, the interaction gives the improvement effect directly and the $\beta_1 + \beta_2$ test is uninformative - maturity is the one term where lenders respond to gains but not to losses. For covenants both directions

are statistically indistinguishable. The data is consistent with lenders treating equivalent gains and losses evenly. Thus, Hypothesis 2 is not supported. Lenders do adjust some non-price conditions of the contract, especially collateral with the change of ESG performance, however, leaving price terms unaffected. When it comes to intensification of the change with the deterioration or improvement of ESG, it appears that they treat gains and losses evenly rather than penalizing deterioration more heavily.

In table 5 the results for Hypothesis 3 are presented.

Table 5. H3 - E, S, G pillar scores and tests of equality across pillars

	Spread	Amount	Maturity	Collateral	Covenants
main					
Standardized values of escore	-0.0339** (0.0166)	-0.00733 (0.0365)	0.441 (0.677)	0.0152 (0.167)	-0.154 (0.108)
Standardized values of sscore	-0.0269* (0.0161)	0.00287 (0.0380)	-0.914 (0.684)	-0.204 (0.172)	0.00643 (0.115)
Standardized values of gscore	-0.0163 (0.0117)	-0.000896 (0.0231)	0.401 (0.443)	0.0248 (0.110)	0.0860 (0.0703)
b(E) - b(S)	-0.0070	-0.0102	1.3555	0.2191	-0.1609
SE of difference	0.0283	0.0660	1.1755	0.2857	0.1997
p: E = S	0.805				
Observations	2644	2644	2644	2551	2619
R-squared	0.557	0.545	0.397		
Pseudo R-squared				0.478	0.137
Borrower clusters	609	609	609	588	600

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

Table 5 reports estimates using pillar scores, so that coefficients are comparable across pillars and represent the effect of a one standard deviation increase (Wald tests of equality between the E, S and G pillar coefficients for H3 are provided in Appendix, tables A9). On loan spreads, both the pillars “E” and “S” are individually significant and have a negative sign, while “G” is

not. The results on this stage are consistent with Hypothesis 3, with the environmental pillar carrying the largest coefficient, followed by the social and then the governance pillar. However, the linear restriction tests provide no evidence of differential pricing across the pillars. The estimated gap between “E” and “S” is rather small relative to its standard error, therefore equality cannot be rejected at conventional levels. Thus, Hypothesis 3 is therefore not supported. While environmental and social performance are each associated with lower borrowing costs, the results do not allow us to conclude that lenders weigh one pillar more than another.

In tables 6 and 7 the results for Hypothesis 4 are presented.

Table 6. H4a - Environmental score x CSR disclosure quality

	Spread	Amount	Maturity	Collateral	Covenants
escore	-0.00182*** (0.000688)	-0.00106 (0.00116)	0.0119 (0.0261)	0.00201 (0.00590)	-0.00266 (0.00383)
Environmental score x High- disclosure dummy	0.000869 (0.000906)	0.00146 (0.00170)	0.00206 (0.0377)	-0.0155 (0.00987)	-0.00303 (0.00568)
=1 if CSR strategy score > annual median	-0.0775 (0.0506)	-0.0591 (0.101)	-1.585 (2.220)	0.758 (0.517)	0.105 (0.346)
Observations	2645	2645	2645	2552	2620
R-squared	0.544	0.544	0.397		
Borrower clusters	610	610	610	589	601

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

Table 7. H4b - Social score x CSR disclosure quality (robustness)

	Spread	Amount	Maturity	Collateral	Covenants
main sscore	-0.00184** (0.000779)	-0.00231 (0.00153)	-0.0309 (0.0314)	-0.00124 (0.00762)	-0.000122 (0.00507)
Social score x High-disclosure dummy	0.0000306 (0.000948)	0.00462** (0.00203)	0.0332 (0.0418)	-0.0212** (0.0107)	-0.00311 (0.00672)

=1 if CSR strategy score > annual median	-0.0390 (0.0580)	-0.247** (0.126)	-2.997 (2.654)	1.186** (0.599)	0.0756 (0.436)
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Observations	2652	2652	2652	2559	2627
R-squared	0.544	0.545	0.396		
Borrower clusters	609	609	609	588	600

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

There is no evidence disclosure quality changes the strength of the ESG-pricing relationship, in the E-score model. However, for the “S” some interesting dynamics is observed. In social dimension disclosure quality moderates non-price terms rather than price: amount and collaterals. The most likely explanation for such difference is that environmental performance relies heavily on hard, quantitative, externally observed data (emissions, energy, waste). Whereas social performance of a firm is based on softer, self-reported and it is much harder to verify indicators (labor practices, community, workforce). Therefore, banks-lenders feel a need for a credibility filter for “S” but not really for “E”. That is exactly the observed pattern: no moderation in Table 6, moderation in Table 7.

In table 8 and 9 the results for Hypothesis 5 are presented.

Table 8. H5.1 - ESG x Time period

	Spread	Amount	Maturity	Collateral	Covenants
esg	-0.00318*** (0.00108)	-0.00162 (0.00214)	0.0219 (0.0440)	-0.0141 (0.0114)	-0.00932 (0.00625)
ESG x Post-2015 (Paris Agreement) dummy	0.000296 (0.00112)	0.00155 (0.00212)	-0.0455 (0.0473)	0.00991 (0.0104)	0.0104 (0.00691)
=1 if year >= 2015	-0.135** (0.0595)	0.0212 (0.104)	0.190 (2.457)	-0.0720 (0.486)	-1.120*** (0.379)
Observations	2661	2661	2661	2568	2635
R-squared	0.521	0.538	0.356		
Borrower clusters	610	610	610	589	601

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

There is no evidence that the ESG-loan terms relationship differs between the pre- and post-2015 periods. The way banks provide loans in response to ESG performance looks the same before and after the Paris Agreement. However, borrowers obtained cheaper and less restrictive loans in the later period, but this shift applied uniformly and did not reward specifically stronger ESG performers. The market moved, while the ESG gradient stayed where it was. This stability suggests that banks were already incorporating ESG considerations into their lending decisions well before the 2015 policy turning point, which suggests lenders had been treating ESG performance as an indicator of credit risk already, rather than accepting it as a new concern triggered by the shift in policies.

Table 9. H5.2 - Emissions_score x Time period

	Spread	Amount	Maturity	Collateral	Covenants
main					
emissions_score	-0.00131** (0.000609)	-0.000343 (0.00144)	0.0271 (0.0260)	-0.00153 (0.00727)	-0.00714** (0.00361)
emissions_score x Post-2015 dummy	-0.000102 (0.000646)	0.000627 (0.00144)	-0.0392 (0.0261)	-0.000678 (0.00653)	0.00458 (0.00395)
=1 if year >= 2015 (post-Paris Agreement), 0 if 2010-2014	-0.131*** (0.0349)	0.0672 (0.0692)	-0.348 (1.393)	0.309 (0.319)	-0.793*** (0.234)
Observations	2645	2645	2645	2552	2620
R-squared	0.529	0.539	0.357		
Borrower clusters	610	610	610	589	601

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

If we choose to take only pillar “E” into account during estimation, the hypothesis 5 still remains unsupported (table 9).

In table 10 the results for Hypothesis 6 are presented.

Table 10. H6 - ESG x Firm size

	Spread	Amount	Maturity	Collateral	Covenants
main esg	-0.00237*** (0.000808)	-0.000395 (0.00166)	0.0406 (0.0351)	0.000336 (0.00861)	0.00455 (0.00560)
ESG x Above- median firm size dummy	-0.00176 (0.00111)	-0.000374 (0.00228)	-0.112** (0.0482)	-0.0129 (0.0108)	-0.0120* (0.00719)
=1 if firm size > annual median	0.105 (0.0687)	-0.0351 (0.138)	4.476 (2.972)	0.108 (0.670)	1.031** (0.452)
Observations	2661	2661	2661	2568	2635
R-squared	0.546	0.544	0.397		
Borrower clusters	610	610	610	589	601

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

The hypothesis 6 that states that bigger firms get a bigger spread discount from ESG is not statistically confirmed. The results may look non-intuitive, but the following explanation can be offered. For above-median borrowers, stronger ESG performance is associated with both shorter maturities and fewer covenants - and since covenant protection and maturity are jointly determined and tend to move together (Billett et al., 2007), this represents a single, uniformly less restrictive contractual package rather than two unrelated effects. Although a shorter maturity might at first appear to be a less favorable term, it is not restrictive in this context: banks impose long horizons together with tighter monitoring, so a shorter, covenant-light facility represents lighter rather than heavier contractual control. One of interpretations is that large, transparent borrowers face few refinancing constraints and consequently draw on bank facilities for short-term liquidity while funding long-horizon investment in public markets, rather than a constraint imposed by lenders.

In tables 11 and 12 the results for Hypothesis 7 are presented.

Table 11. H7.1 - ESG x High-polluting sector

	Spread	Amount	Maturity	Collateral	Covenants
esg	-0.00293***	0.000745	-0.0281	0.00703	-0.000405

	(0.000705)	(0.00157)	(0.0319)	(0.00688)	(0.00453)
ESG x High-polluting sector dummy	-0.000967 (0.00113)	-0.00295 (0.00253)	0.0199 (0.0448)	-0.00451 (0.0110)	-0.00125 (0.00671)
=1 if high polluting / environmentally sensitive sector	0.00662 (0.0653)	0.183 (0.137)	-2.337 (2.715)	0.173 (0.525)	-0.0462 (0.372)
Observations	2661	2661	2661	2661	2661
R-squared	0.491	0.516	0.371		
Borrower clusters	610	610	610	610	610

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

We find no evidence in this sample that the ESG-loan terms relationship is any stronger or weaker in high-polluting sectors than elsewhere. However, the analysis was repeated separately just for the “E” pillar, as it is the most representative for the current hypothesis (table 12).

Table 12. H7.2- “E” pillar x High-polluting sector

	Spread	Amount	Maturity	Collateral	Covenants
main					
E pillar score	-0.00170*** (0.000499)	0.000676 (0.00107)	-0.00860 (0.0201)	-0.000571 (0.00449)	-0.00406 (0.00309)
E pillar x High-polluting sector dummy	-0.00119 (0.000854)	-0.00296 (0.00185)	0.00219 (0.0342)	0.00465 (0.00941)	0.000603 (0.00506)
=1 if high-polluting / environmentally sensitive sector	0.0106 (0.0498)	0.181* (0.100)	-1.131 (2.079)	-0.261 (0.418)	-0.153 (0.272)
Observations	2645	2645	2645	2645	2645
R-squared	0.500	0.517	0.372		
Borrower clusters	610	610	610	610	610

Note: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: own elaboration based on data from Refinitiv and DealScan

The coefficient on spread is significant at 1% just like on the composite level. The hypothesis 7 is still not supported, however it can be concluded that most significant part of the ESG discount is based on E pillar.

5. Conclusions

The study investigates whether US banks modify their syndicated loan conditions based on ESG rating scores of the firm borrowers. The research is motivated by the following gaps: reliance of majority of previous research on composite scores instead of decomposing the pillars and then estimating them jointly; and investigation of purely one loan term, usually the spread.

Current thesis is based on Dealscan + LSEG database, covering 2,944 loan tranches, 589 U.S. listed firms and time period from 2010 to 2025. This database is used to examine simultaneously following loan characteristics in the framework of the thesis topic: spread, amount, maturity, collateral, covenants.

The research tested 7 different hypotheses. Hypothesis 1 was partially supported – ESG was found to be priced only in the spread. No association with amount, maturity, collateral or covenants is observed. Hypothesis 2 is not supported. There is a symmetrical response of collateral to gains and losses, while maturity only responds to improvements, the effect from deterioration is not detectable. Price terms were shown to have no response in both directions. Moving to hypothesis 3, no evidence of different pillar weighting was found. Hypothesis 4 showed asymmetric results for pillars “E” and “S”. For pillar “E” disclosure gap does not moderate the relationship with pricing, while for “S” pillar it moderates amount-pillar and collateral-pillar relationships but not spread. As for hypothesis 5 and 7, we find that there are no period or sector effect from the ESG rating score change. However, the loans became uniformly cheaper and possessing fewer heavy covenants post-2015. Hypothesis 6 is not confirmed in its original form, however, size does affect two non-price conditions - the ESG gradient by maturity and by the frequency of covenant use significantly differs between borrowers above and below the median.

Moving to the relationship between the findings and previous findings, the results confirm the spread results and the direction of the ESG discount from Goss & Roberts (2011). In general, the findings of the research contradict the studies that report the effects of ESG on amount, maturity and collateral, probably because they are examined separately without conditioning the rest. Results of the research support findings of Carnevale & Drago (2024). The investigation confirms that lenders respond to specific information rather than being responsive to composite scores. The results of the thesis research diverge from work reporting and intensification of ESG pricing after the Paris Agreement. The ESG gradient is statistically indistinguishable before and after 2015: the later period brought cheaper and less restrictive loans for all borrowers, but no additional reward for sustainability performance.

This thesis has important implications for various categories of stakeholders. Firstly, for borrower companies. ESG indicators are not limited to reporting and interaction with investors; they also influence credit relationships and are reflected in the conditions offered by banks. This means that executives should consider sustainability as one component of debt cost rather than a separate reputation exercise. Secondly, for banks as lenders. Stronger ESG performance is associated with lower spreads even after controlling standard credit risk factors, but the evidence does not support weighing any one dimension more heavily than another: the environmental, social and governance pillars are not priced differently. It also raises a question for banks: if sustainability risk is reflected in price but not in contractual guarantees, then risk is compensated but not necessarily mitigated. Thirdly, for rating providers. The informational value sought by lenders lies in decomposed and comparable indicators. Composite scores seem to have limited weight in the contracting process. This provides an argument for greater methodological transparency and disclosure at the individual component level.

The current research possesses some gaps that were not covered in the analysis as follows. First, only large listed firms with Refinitiv coverage are analyzed. Therefore, private or middle-

market firms are not covered, and findings cannot be applied to them as they are functioning in totally different market conditions. This suggests that future research can expand the exploration of the topic to private or medium sized firms. Second, the research only covers syndicated loans. However, group of banks are not the only possibility of lender that a firm-borrower can go to bilateral bank lending, bonds and private credit opportunities are not investigated in the research. Application of the research question to the other types of the financing could be analyzed in future research. Third, only US-borrowers are studied, therefore no non-U.S. borrowers are present. That makes it impossible to extend the following studies on the other countries with different economic, banking and credit conditions. Fourth, the influence of ESG-change on various loan type and purpose of the loans are not investigated, despite the fact that they are available in Dealscan but not used. That also gives an opportunity for future research, which can test the hypothesis on diversified loans.

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APPENDIX

Table A1. Dependent variables

Variable Name	Type	Source
Dependent variables		
All-In Spread Drawn	Dependent	DealScan
Loan Amount	Dependent	DealScan
Loan Maturity	Dependent	DealScan
Collateral (Secured)	Dependent	DealScan
Covenants	Dependent	DealScan

Source: own elaboration based on data from DealScan

Table A2. Explanatory variables

Variable Name	Type	Hypothesis	Source
ESG Score	Explanatory	H1, H5, H6, H7	Refinitiv
Δ ESG (Change in ESG)	Explanatory	H2	Refinitiv, computed
ESG Increase Dummy	Explanatory	H2	Computed
Environmental Score (E)	Explanatory	H3, H4	Refinitiv
Social Score (S)	Explanatory	H3, H4 (robustness)	Refinitiv
Governance Score (G)	Explanatory	H3	Refinitiv
Emissions Score	Explanatory	H3 (category level)	Refinitiv
Resource Use Score	Explanatory	H3 (category level)	Refinitiv
Environmental Innovation Score	Explanatory	H3 (category level)	Refinitiv
Workforce Score	Explanatory	H3 (category level)	Refinitiv
Human Rights Score	Explanatory	H3 (category level)	Refinitiv
Community Score	Explanatory	H3 (category level)	Refinitiv
Product Responsibility Score	Explanatory	H3 (category level)	Refinitiv
Management Score	Explanatory	H3 (category level)	Refinitiv

Shareholders Score	Explanatory	H3 (category level)	Refinitiv
CSR Strategy Score	Explanatory	H3 (category level)	Refinitiv
CSR Dummy (above annual median)	Explanatory	H4	Refinitiv, computed
Time Period Dummy	Explanatory	H5	Computed
Firm Size (interaction)	Explanatory	H6	Refinitiv
High-Polluting Industry Dummy	Explanatory	H7	Refinitiv (SIC), computed

Sources: own elaboration based on data from Refinitiv

Table A3. Control variables

Variable Name	Type	Source
Control variables (firm level)		
Leverage	Control	Refinitiv
ROA (Return on Assets)	Control	Refinitiv
Interest Coverage	Control	Refinitiv
Firm Size	Control	Refinitiv
Market-to-Book Ratio	Control	Refinitiv
Tangibility	Control	Refinitiv
Control variables (loan level)		
Loan Size	Control	DealScan
SLL Dummy (KPI-linked)	Control	DealScan, computed

Sources: own elaboration based on data from Refinitiv and DealScan

Table A4. Moderating variables

Variable Name	Type	Source
Moderating variables		

CSR Dummy2(above/below median)	Moderator (H4)	Refinitiv, computed
Time Period Dummy	Moderator (H5)	Computed
Firm Size (interaction)	Moderator (H6)	Refinitiv
High-Polluting Industry Dummy	Moderator (H7)	Refinitiv (SIC code), computed

Source: own elaboration based on data from Refinitiv

Table A5. Fixed effects

Variable Name	Type	Source
Fixed effects		
Year Fixed Effects	Fixed Effect	Computed
Sector Fixed Effects	Fixed Effect	Refinitiv (SIC code)

Source: own elaboration based on data from Refinitiv

² The CSR transparency dummy variable appears in both Table 2 and Table 4. In Hypothesis 4 it plays a role of the moderating variable capturing ESG disclosure quality, whereas in Hypotheses 5-7 it functions as a dummy variable. Its inclusion in both tables is needed due to this its dual role.

Table A6. The correlation matrix of the variables (Columns 1-12)

Variable	(1) ln_spread	(2) ln_amount	(3) Tenor/Maturity (month)	(4) esg	(5) escore	(6) sscore	(7) gscore	(8) emissions_score	(9) resource_use	(10) env_innovation	(11) workforce	(12) hr
(1) ln_spread	1											
(2) ln_amount	-0.305***	1										
(3) Tenor/Maturity (month)	0.329***	-0.227***	1									
(4) esg	-0.405***	0.419***	-0.300***	1								
(5) escore	-0.376***	0.405***	-0.280***	0.881***	1							
(6) sscore	-0.387***	0.408***	-0.287***	0.903***	0.755***	1						
(7) gscore	-0.223***	0.199***	-0.169***	0.633***	0.379***	0.356***	1					
(8) emissions_score	-0.361***	0.418***	-0.285***	0.817***	0.872***	0.732***	0.375***	1				
(9) resource_use	-0.370***	0.406***	-0.287***	0.855***	0.888***	0.787***	0.370***	0.807***	1			
(10) env_innovation	-0.254***	0.173***	-0.155***	0.490***	0.642***	0.348***	0.223***	0.345***	0.379***	1		
(11) workforce	-0.335***	0.413***	-0.249***	0.792***	0.692***	0.825***	0.343***	0.714***	0.725***	0.250***	1	
(12) hr	-0.311***	0.296***	-0.228***	0.709***	0.598***	0.816***	0.252***	0.551***	0.652***	0.301***	0.530***	1
(13) community	-0.315***	0.353***	-0.252***	0.702***	0.622***	0.732***	0.298***	0.622***	0.618***	0.292***	0.634***	0.459***
(14) product_response	-0.278***	0.272***	-0.183***	0.625***	0.516***	0.710***	0.232***	0.485***	0.511***	0.291***	0.491***	0.431***

(15) managem ent	-0.143***	0.0963***	-0.0899***	0.468***	0.203***	0.195***	0.937***	0.187***	0.194***	0.136***	0.202***	0.124***
(16) sharehold ers	-0.0814***	0.127***	-0.0942***	0.273***	0.153***	0.139***	0.483***	0.190***	0.147***	0.0793***	0.128***	0.0498**
(17) csr	-0.368***	0.384***	-0.310***	0.777***	0.761***	0.705***	0.436***	0.763***	0.763***	0.399***	0.633***	0.602***
(18) tdt	0.173***	0.0761***	-0.00207	0.0169	0.0397**	0.00883	-0.00798	0.0578***	0.0472**	-0.0566***	0.00836	0.0196
(19) netincome ta	-0.275***	0.0731***	-0.00985	0.0907***	0.0983***	0.0916***	0.0293	0.0611***	0.0903***	0.0813***	0.0690***	0.0801***
(20) ebitinteres t	-0.211***	-0.0401**	0.0384**	-0.0526***	-0.0706***	-0.0119	-0.0629***	-0.0775***	-0.0477**	-0.0564***	-0.0214	-0.00441
(21) mtb	-0.152***	0.0571***	0.0240	0.0485**	0.0621***	0.0653***	-0.0402**	0.0501***	0.0775***	0.0385**	0.0895***	0.0599***
(22) ppeta	0.0782***	-0.0492**	-0.0513***	-0.00716	0.0606***	-0.0861***	0.100***	0.139***	0.0132	-0.0571***	-0.0563***	-0.120***
(23) ln_ta	-0.391***	0.676***	-0.389***	0.588***	0.578***	0.542***	0.305***	0.612***	0.556***	0.274***	0.565***	0.351***

Notes: Pearson correlations. N = 2,614 (listwise deletion). * p < 0.10, ** p < 0.05, *** p < 0.01.

Source: own elaboration based on data from Refinitiv and DealScan

Table A7. Pairwise Correlations (Columns 13–23)

Variable	(13) community	(14) product resp	(15) management	(16) shareholders	(17) csr	(18) tdta	(19) netincometa	(20) ebitinterest	(21) mtb	(22) ppeta	(23) ln ta
(1) ln_spread											
(2) ln_amount											
(3) Tenor/Maturity (month)											
(4) esg											
(5) escore											
(6) sscore											
(7) gscore											

(8) emissions_score												
(9) resource_use												
(10) env_innovation												
(11) workforce												
(12) hr												
(13) community	1											
(14) product_resp	0.376***	1										
(15) management	0.160***	0.120***	1									
(16) shareholders	0.127***	0.133***	0.235***	1								
(17) csr	0.590***	0.439***	0.203***	0.166***	1							
(18) tdta	0.00958	-0.0308	-0.0126	0.0151	-0.000408	1						
(19) netincometa	0.0814***	0.0662***	0.0145	-0.00178	0.0791***	-	1					
(20) ebitinterest	-0.0195	-0.00366	-0.0342*	-0.0585***	-	0.0760***	-0.353***	0.493***	1			
(21) mtb	0.0524***	0.00963	-0.0358*	-0.0607***	0.0900***	0.0264	0.0720***	0.216***	0.130***	1		
(22) ppeta	-0.00212	-0.0673***	0.0620***	0.116***	0.0907***	0.125***	-0.113***	-0.136***	-0.145***	1		
(23) ln ta	0.486***	0.379***	0.152***	0.203***	0.565***	0.0858***	-0.0555***	-0.149***	-0.0228	0.135***	1	

Source: own elaboration based on data from Refinitiv and DealScan

Table A8. Linear restriction tests for all five contractual terms for H2

	Spread	Amount	Maturity	Collateral	Covenants
Deterioration effect b_1	-0.00671	-0.00680	-0.516	0.226*	-0.100
Improvement effect b_1+b_2	-	-	-	-0.288	-
SE of b_1+b_2	-	-	-	0.088	-
p: $b_1+b_2 = 0$	-	-	-	0.001	-
p: $b_2 = 0$ (arms differ)	0.267	0.599	0.034	0.003	0.740
p: $2b_1+b_2 = 0$ (symmetry)	-	-	-	0.587	-
AME, deterioration	-	-	-	0.0058	-0.0181
AME, improvement	-	-	-	-0.0065	-0.0149
Observations	491	491	491	392	451
Firm clusters	185	185	185	150	164

Notes: Tests estimated after the models reported in Table 4. b_1 is the effect of a one-point ESG deterioration; b_1+b_2 is the effect of a one-point improvement, recovered by a linear restriction test. The symmetry restriction $2b_1+b_2 = 0$ tests whether improvements and deteriorations of equal size have effects of equal magnitude and opposite sign. Columns 4 and 5 are logit models; average marginal effects (AME) are expressed in probability points. Em dashes indicate restrictions that were not estimated for that outcome. Standard errors clustered at the borrower level.

Source: own elaboration based on data from Refinitiv and DealScan

Table A9. Wald tests of equality between the E, S and G pillar coefficients for H3

	Spread	Amount	Maturity	Collateral	Covenants
b(E) – b(S)	–0.0070	–0.0102	1.3555	0.2191	–0.1609
SE of difference	0.0283	0.0660	1.1755	0.2857	0.1997
p: E = S	0.805	0.877	0.249	0.443	0.420
p: E = G	0.398	0.893	0.961	0.964	0.070
p: S = G	0.640	0.936	0.134	0.292	0.570
p: E = S = G (joint)	0.659	0.988	0.324	0.573	0.153
Observations	2644	2644	2644	2551	2619
Firm clusters	609	609	609	588	600

Notes: Wald tests of the null that the pillar coefficients are equal, estimated after the models reported in Table 5. Pillar scores are standardised, so coefficients express the effect of a one standard deviation increase. Columns 4 and 5 are logit models. Year and 2-digit SIC fixed effects, full controls, and standard errors clustered at the borrower level are included throughout.

Source: own elaboration based on data from Refinitiv and DealScan